IoT based System for Sleep Apnea Detection and Disorder Measurement

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Abstract **Sleep apnea is a critical condition which is identified by sleep disturbances caused by interrupted breathing, often accompanied by low oxygen levels, fatigue, and cardiovascular complications. Polysomnography is the conventional method for diagnosing sleep apnea that is expensive since it requires supervision in a clinical setting, and is impractical for continuous home monitoring. In this paper we propose a continuous, real time Internet of Things (IoT) based health monitoring system designed to detect sleep apnea using a threshold detection method. This system utilizes different physiological sensors like MAX30102 to measure heart rate and blood oxygen saturation (SpO2), a respiratory effort sensor to monitor breathing patterns, a galvanic skin response (GSR) sensor to show stress levels and an accelerometer to detect sleeping posture. These sensors are connected to an ESP32 microcontroller, which processes the data from the sensors and transmits it via Wi Fi to the Thing Speak cloud platform. When the system encounters any sensor reading that exceeds the predefined threshold value, it automatically triggers alerts using cloud services. This method does not include complex AI models, is noninvasive, cost effective, customizable, and permits user driven monitoring. As a result it offers a promising solution for at home sleep apnea detection, which is not only accessible but also enables patients and healthcare providers to respond immediately to concerning health events.**

Keywords ***Sleep apnea detection, IoT, wearable sensors, real time alert system, home based monitoring.***

I. INTRODUCTION

Sleep apnea [1] is a well-known yet serious sleep disorder in which breathing repeatedly stops and restarts, which causes oxygen desaturation, fragmented sleep, and severe health complications: cardiovascular disease, hypertension, and cognitive decline [2]. The disorder divides into Obstructive Sleep Apnea (OSA) due to relaxation of throat muscles and Central Sleep Apnea (CSA), whereby the brain does not send correct signals to the respiratory muscles [5]. Millions suffer from it worldwide, with many undiagnosed due to the constraints of typical diagnostic methods [6]. Polysomnography (PSG), where the subject stays overnight in the sleep lab to measure several physiological parameters such as blood oxygen levels, respiratory effort, and brain activity, is the gold standard for the diagnosis of sleep apnea [7]. However, due to its cost, it requires specialized medical supervision, and, therefore, cannot support continuous long term monitoring in the home setting [8]. As a result, there is a greater demand for real time, home based sleep apnea detection systems that allow continuous monitoring and early intervention. Thanks to the advances in IoT technology, wearable, non -invasive sleep monitoring solutions are developed with real time physiological data collection and detection of the apnea events [9]. Such IoT- based approaches include the use of machine learning based models which analyze sensor data in order to classify sleep apnea events [10]. However, these methods usually require high computational power and large datasets for training, making it a challenge for achieving real time processing in low power wearable devices [11].This proposed paper presents a threshold respiratory effort, and accelerometer sensors coupled with an ESP32 microcontroller for the real time monitoring based IoT system for sleep apnea detection which is not done with the use of machine learning but predefined physiological thresholds used to detect apneic events. It incorporates SpO_2, GSR, of key physiological parameters [12]. Detected data is constantly compared against the predefined thresholds, enabling instant alerts through SMS and cloud notifications to caregivers or medical professionals in case of the emergence of any abnormality. There is thus low latency detection, less power consumption and real time operation, as compared to traditional AI based models.

II. RELATED WORKS

Polysomnography (PSG) is currently the most trusted method for diagnosing sleep apnea because it measures a wide range of body functions during sleep—like brain waves, heart activity, oxygen levels, breathing patterns, and muscle movements. Research such as [4] shows that it can identify sleep apnea with over 95% accuracy. But while it's highly effective, it comes with several drawbacks. For one, it's expensive and requires patients to spend the night in a specialized sleep lab, which isn't ideal for everyday screening. The many wires and sensors can also make sleeping uncomfortable, potentially affecting the results. Plus, since it only records data from a single night, it might miss variations in sleep patterns that occur over time. On top of that, the data collected needs to be reviewed by trained professionals, which adds to the cost and waiting time for a diagnosis. To make sleep apnea detection more accessible, researchers are turning to artificial intelligence (AI) and machine learning. These systems use data from wearable sensors, like heart rate and oxygen monitors, to detect apnea events automatically. One study in [5] trained a deep learning model on ECG and SpO_2 data and achieved high accuracy. However, such models typically need a lot of computing power, making them less practical for real time use on everyday devices. Other approaches, like smartphone based systems using motion and heart rate sensors, offer more convenience but can generate false alarms due to movement or sensor errors. AI also comes with challenges like needing large, labeled datasets for

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training, depending on cloud computing (which raises privacy concerns), and draining battery life due to constant data processing. At the same time, IoT (Internet of Things) devices have been used to create wearable systems that monitor sleep more comfortably at home. For example, [5] describes a system that tracks oxygen levels and breathing effort and sends alerts when levels drop too low. However, it sometimes triggers false alarms because it doesn't cross check multiple signals. Another system in [6] added more sensors like those for airflow, movement, and ECG—to improve accuracy, but wearing ECG patches can still be uncomfortable for long term use. Common issues with IoT systems include limited data sources, reliance on cloud servers (which causes delays and uses more power), and discomfort due to bulky or wired components.

To overcome these problems, some researchers have started combining AI and IoT in hybrid systems. These setups aim to use smart, wearable devices that can process data locally or send it to the cloud for more advanced analysis. One such system in [7] used sensors for oxygen levels, airflow, and body position alongside a deep learning model hosted on the cloud. It improved detection accuracy but had latency issues due to data transfer times. Another example from [8] featured a smart bed with built in sensors and on device processing, which helped reduce cloud reliance but wasn't portable or widely usable. While these hybrid systems show promise by blending accuracy with convenience, they still face hurdles like high hardware costs, limited customization for individual users, and ongoing dependence on internet connectivity. Overall, the goal is to create a solution that's accurate, comfortable, energy efficient, secure, and suitable for daily home use—but getting there remains a work in progress.

III. PROPOSED SYSTEM

The current sleep apnea detection techniques are usually expensive, invasive, necessitate a visit to a clinical practitioner for diagnosis, or, in cases where AI based approaches are adopted, require considerable computing power. A novel IoT based, threshold driven wearable system is proposed in this work for continuous monitoring based at home, minimally uncomfortable and consuming less power. Our approach is a noninvasive, wearable alternative: uninterrupted physiological monitoring avoiding clinical supervision, unlike PSG, the gold standard for sleep apnea diagnosis, requiring lengthy stays at hospitals, multiple wired sensors, and trained personnel monitoring the data analysis [8]. On the other hand, AI based techniques, such as deep learning model based apnea detection, are lesser feasible for real time processing at home environments because of their dependence on huge datasets and substantial computing power [9]. The PI system, meanwhile, deposits real time apnea detection through threshold detection involving continuous comparison of physiologic data from sensors against physiological thresholds defined before commencing the clinical and experimental model. This is illustrated in Figure 1. This system consists of a multi sensor integration module linked to an ESP32 microcontroller to gather oxygen saturation (SpO2), galvanic skin response (GSR), and respiratory effort and accelerometer data in order to identify incidents of sleep apnea. The SpO2 sensor is continuously tracking blood oxygen levels and detecting drops below 90%, a clinically significant level and a primary indicator of oxygen desaturation from obstructive sleep apnea (OSA)[10]. The GSR sensor measures skin conductance levels, which are affected by the sympathetic nervous system response during the apnea episodes and lead to increased sweating and stress reactions [11]. The respiratory effort sensor monitors thoracic and abdominal

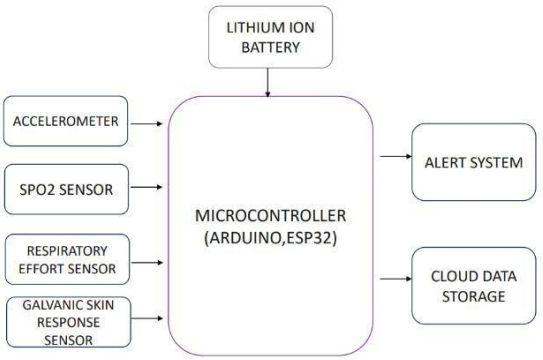


Fig 1. Block Diagram of the Proposed System

movement and is able to identify breathing irregularities and airflow blocks that help indicate central or obstructive sleep apnea [12]. The accelerometer tracks body movement and posture as certain positions while sleeping such as lying supine can worsen airway collapse and increase apnea severity [13]. By means of appropriate thresholds set based on past clinical studies for the above four physiological parameters, apnea events can be recognized in real time by the system, and this initiates the immediate alarm to caregivers and healthcare providers.

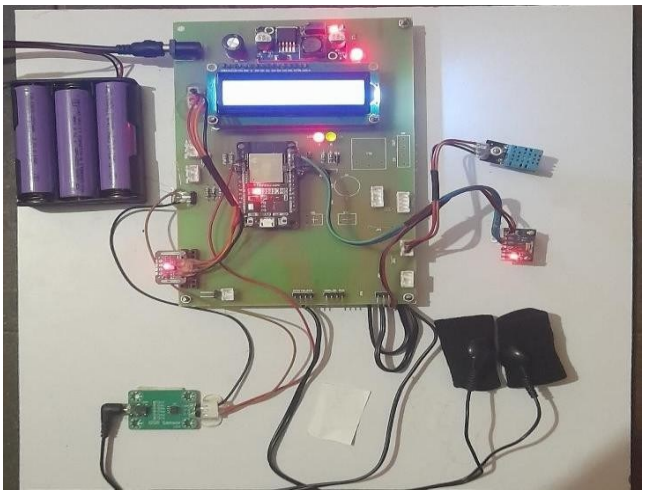


Fig 2. The Proposed Hardware Device

Our device depicted in Figure 2 is not impeded by typical IoT systems requiring fixed position sensors that cannot be used on patients with commonly changing sleeping positions. In contrast, this new wireless system provides massive flexibility that allows for portable usability. In contrast with AI driven analysis of sleep apnea, like the MLA ones trained on extensive PSG datasets, they also suffer notability from false positives, privacy issues, and energy drain that make them practically unfeasible for real time, battery powered wearable applications. On the contrary, the lightweight, threshold-based algorithm for our setup is meant to provide instantaneous detection of apnea without dramatic reliance on high performance cloud computing, thus minimizing latency and power dissipation. The ESP32 microcontroller acts as the major CPU of the system, involve efficient handling of data acquisition and sensor formats and usages at the same in contemporaneous Wi Fi connection to a cloud server. The system immediately sends a warning of the event through SMS, email, and a mobile application to conclude alerting caregivers and medical personnel to tell them the importance of fast medical

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aid so as to avert severe hypoxic events. Other technologies in wearables include rudimentary SpO2 monitoring as shown in Figure 3 in commercial smartwatches and fitness bands, but these largely lack multisensory integration and sometimes leave much in detailed respiratory effort analysis required by the clinicians in sleep apnea identification. Our system incorporates multisensory fusion methods that specially provide high precision, reduce false alarms, and increase sensitivity to apnea related physiologic changes, besides a regular bandwidth of remote access for physicians into elevating or lowering sensor thresholds that pertain to specific patient complications. Comparatively with regular PSG setups in which manually sourced overnight recordings analyze, the system accords automatic detection and alert notifications in real time, greatly lightening the workload for health facilities while exhibiting proactive measures to keep sleep apnea symptoms at bay. This system integrates various cost effective IoT components that provide real time processing, wireless connectivity, and multi sensor data fusion, hereby advancing reliable, scalable, and efficient alternatives to the current challenges of sleep apnea diagnostics. The outcome is envisaged with factors like better early detection rates, reduced costs in healthcare, and the possibility of home-based monitoring, especially with the elderly, those inconveniently located, and high-risk persons needing continued apnea surveillance.

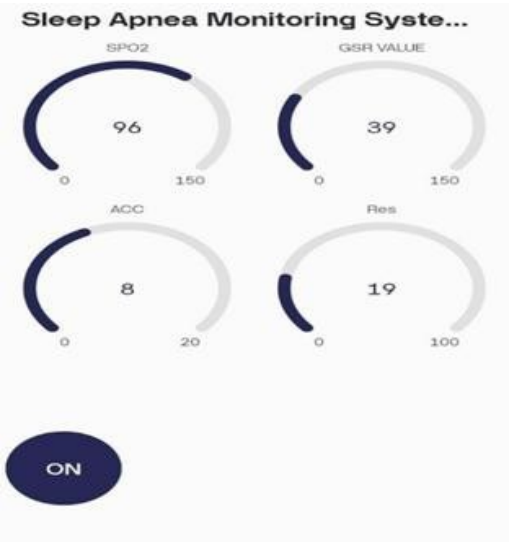


Fig 3. Monitoring of Sleep Apnea through various sensors

IV. METHODOLOGY

The system implemented here is a lightweight, rule-based detection method that uses pre-defined clinically verified threshold values as a basis for Apnea detection, unlike the much more complex AI based models that require great volumes of data along with complex algorithms for training. The detection algorithm indeed follows these steps:

Step 1: The SpO2 drop indicator: when SpO2 is below 90% for more than 10 seconds it is recognized as a possible apnea event.

1) Respiratory effort monitoring when no thoracic or abdominal motion is evident, in other words, 10 seconds have passed with no motion Indicates cessation of breath
2) GSR and stress analysis The GSR exceeds specific threshold values; thus, they trigger an alarm at the higher levels stress is caused from the activation of the sympathetic system initiated by apnea
3) Based on the body position the Apnea event probability value weighted becomes high if the accelerometer indicates that a certain position is supine the airway obstruction statement therefore supports itself The confirmation of one such event Instant Alert notification generation occurs . 5) If there exist multiple parameters fulfilling apnea getting conditions simultaneously. Section D deals with wireless communication and alert system After detecting apnea events, the ESP32 relays data via WiFi to a cloud-based platform, thus enabling real time monitoring and giving access to doctors for historical apnea events from afar.

The alerting mechanism contains modalities such as:

1) SMS notifications: The real time SMS and Email alerts should reach the caregivers and doctors to ensure a timely intervention.
2) Cloud based dashboard: A Cloud based monitoring dashboard provides Apnea records, showing the trends for doctor analysis.
3) Notifications on a mobile phone real time alerts are sent through a smartphone app to the users for immediate awareness and possible self-correction like changing sleep poser. The accuracy of the system built on real time testing with patients suffering from apnea was compared to that of polysomnography, taking care of the three properties under then:

1) The detection accuracy refers to the possibility for the system to detect an apnea episode as associated with a polysomnography result.
2) The false alarm rate is measured as the percentage of apnea events wrongly flagged during the time window measured
3) The user comfort and wearability are given as feedback from the user about the noninvasiveness of the device.
4) The response time refers to the time it takes for detecting an apnea event on issuing an alert. With this, by the collision of the wearable IoT sensors, real time threshold-based detection algorithms, and alert mechanisms into the cloud, they get a cost effective, noninvasive, and efficient home monitoring diagnosis of sleep apnea

V. RESULTS AND DISCUSSION

The suggested IoT based sleep apnea detection system, which is driven by threshold levels, was designed, developed, and tested in a controlled situation to evaluate its effectiveness in detecting and reporting sleep apnea episodes through the noninvasive physiological monitoring of individuals. The system leverages the capabilities of low energy, wearable sensors with a microcontroller designed to be robust (ESP32), as well as real time cloud access via the ThingSpeak platform for visualization and alerts.

The hardware prototype consists of four main sensors: SpO_2 Sensor (MAX30102) – measures blood oxygen saturation and heart rate, with an SpO_2 drop below 90% indicating a hypoxic state and potential for an apnea event. Galvanic Skin Response (GSR) Sensor – measures skin conductivity, which may be related to stress, which may increase during apnea episodes. Respiratory Effort Sensor (Piezoelectric Belt) – records thoracic and abdominal movement for recognizing pauses in breath. Accelerometer (ADXL345) – records motion and body position, useful in correlating sleep posture with apnea severity.

The ESP32 serves as the main device connecting all sensors and

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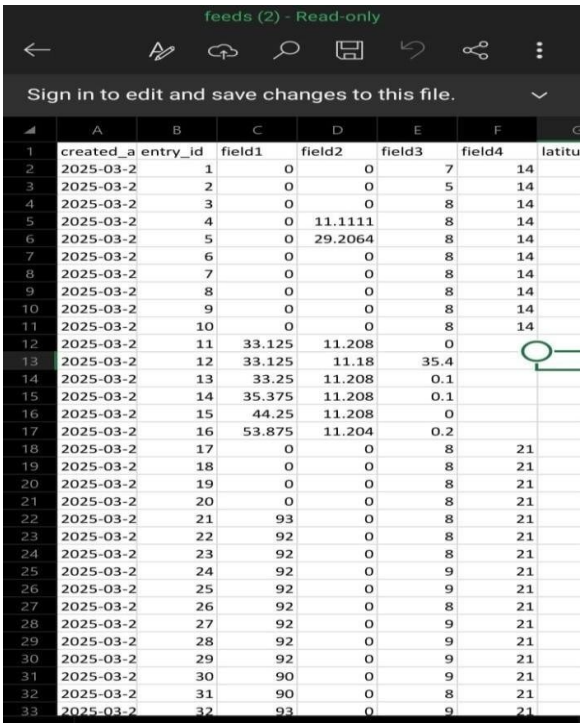
processing the data in real time. The firmware is designed to continually sample sensor values, compare them to preset



Fig 4 Things Speak Platform to view SPO2 and Respiratory data



Fig 5 Things Speak Platform to view GSR and ACC data



	A	B	C	D	E	F	G
1	created_a	entry_id	field1	field2	field3	field4	latitu
2	2025-03-2	1	0	0	7	14	
3	2025-03-2	2	0	0	5	14	
4	2025-03-2	3	0	0	8	14	
5	2025-03-2	4	0	11.1111	8	14	
6	2025-03-2	5	0	29.2064	8	14	
7	2025-03-2	6	0	0	8	14	
8	2025-03-2	7	0	0	8	14	
9	2025-03-2	8	0	0	8	14	
10	2025-03-2	9	0	0	8	14	
11	2025-03-2	10	0	0	8	14	
12	2025-03-2	11	33.125	11.208	0		
13	2025-03-2	12	33.125	11.18	35.4		
14	2025-03-2	13	33.25	11.208	0.1		
15	2025-03-2	14	35.375	11.208	0.1		
16	2025-03-2	15	44.25	11.208	0		
17	2025-03-2	16	53.875	11.204	0.2		
18	2025-03-2	17	0	0	8	21	
19	2025-03-2	18	0	0	8	21	
20	2025-03-2	19	0	0	8	21	
21	2025-03-2	20	0	0	8	21	
22	2025-03-2	21	93	0	8	21	
23	2025-03-2	22	92	0	8	21	
24	2025-03-2	23	92	0	8	21	
25	2025-03-2	24	92	0	9	21	
26	2025-03-2	25	92	0	9	21	
27	2025-03-2	26	92	0	8	21	
28	2025-03-2	27	92	0	9	21	
29	2025-03-2	28	92	0	9	21	
30	2025-03-2	29	92	0	9	21	
31	2025-03-2	30	90	0	9	21	
32	2025-03-2	31	90	0	8	21	
33	2025-03-2	32	93	0	9	21	

Fig 6. Data from Cloud after continuous observation

Figure 4, 5, 6 and 7 show the results. Threshold values, and assess an event for potential apnea.In order to provide remote monitoring and historical tracking, the system is connected to the ThingSpeak IoT cloud platform. The ESP32 will send the processed data using HTTP GET requests to the corresponding ThingSpeak channels where each physiological metric is visualized in real time. The ThingSpeak dashboard can visualize multiple data fields and plot back dated graphs, to provide caregivers and users with a visual of the patient's physiologies during sleep events. For example, the SpO_2 values are plotted continuously, allowing the viewer to visually confirm any trends of desaturation. If oxygen saturation dips below the predefined threshold of 90% and stays below that threshold for a few seconds (e.g., 10 15 seconds), then the event is stored as a possible episode of apnea. Similarly, if the GSR sensor detected an instant increase of skin conductance or if the respiratory effort sensor determines a flatline, then these events are flagged for being incorporated into an alert.

The MATLAB analytics environment on Thing Speak is used to set up automatic detection of abnormal conditions.

Conditional scripts are written to obtain live monitoring of values in real time. When an abnormality is detected such as a sustained drop in SpO_2 or lack of respiratory movement webhooks or Thing HTTPs are triggered. Then they interface with third party services, like IFTTT, to send SMS or email alerts to a caregiver, family member, or physician. The latency that was measured during testing for generating alerts was approximately 2–3 seconds after the detection of the abnormality, allowing for timely notification and response.

As part of the validation, simulated events of apnea were initiated through manipulation of the sensor data inputs and the system was able to respond with accurate detection and alerts reliably. In each trial, the sensor data uploaded automatically to ThingSpeak every 15 seconds, and could be displayed either as a table or a graph. The system was very robust in testing over longer periods of time, running consistently up to 20 hours on a 2200 mAh Li ion battery, if likely enough for an overnight period without recharge needed.

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Regarding data storage, ThingSpeak facilitated the simparchival and export of sensor logs for subsequent offline analysis. Its public and private channel capabilities offered both security and access in line with foundational data privacy considerations. The API key for the channel ensures that only authorized devices are allowed to upload or access data on the system, providing a level of reliability and control. The affordability and simplicity of the system are clear advantages. In contrast to polysomnography systems requiring multiple electrodes, clinical setup, and in person monitoring, the system is wireless, portable, and user configurable for home based monitoring. Furthermore, the absence of any AI/ML components reduces complexity, memory consumption, and power usage, while achieving an acceptable level of accuracy by leveraging logic based on a multi sensor and threshold approach. The other noteworthy benefit of the use of ThingSpeak is its capability for remote threshold customization. The healthcare provider can modify or de novo thresholds by simply writing values to corresponding fields on the ThingSpeak channel, from which the ESP32 retrieves the values during scheduled refreshes. This feature makes the system highly adaptable to various patients based on age, health status, or sleep behavior. In contrast to polysomnography systems requiring multiple electrodes, clinical setup, and in person monitoring, the system is wireless, portable, and user configurable for home-based monitoring. The absence of any AI/ML components reduces complexity, memory consumption, and power usage, while achieving an acceptable level of accuracy by leveraging logic based on a multi sensor and threshold approach. The other noteworthy benefit of the use of ThingSpeak is its capability for remote threshold customization. The healthcare provider can modify or de novo thresholds by simply writing values to corresponding fields on the ThingSpeak channel, from which the ESP32 retrieves the values during scheduled refreshes. This feature makes the system highly adaptable to various patients based on age, health status, or sleep behavior. The sleep position data obtained from accelerometer readings, it is possible to determine position induced apnea. It may then be possible to counsel patients with worsened symptoms while supine to change their sleeping position, adding a preventative dimension to the diagnosis



Fig 7 GSR Data through monitoring

VI. CONCLUSION

The system we developed effectively demonstrates a real time, wearable, and cloud based approach to sleep apnea detection utilizing basic threshold criteria, without using an AI model. It has been formulated to be power efficient, provide cloud syncing, offer rapid alerts, and allow for configurability in remote mode. The use of ThingSpeak further provides a scalable, visual, and customizable monitoring solution. These results verify that such an IoT based solution can be a useful, affordable, and reliable means of assisting in early detection and monitoring of sleep apnea in a home -based environment. The research has addressed issues of a practical and timely nature; which may be important for low resource contexts and for older populations.

REFERENCES

[1] N. Akhtar, “Critical Evaluation of Automated Obstructive Sleep Apnea Detection Using EEG and Machine Learning: Strengths, Limitations, and Future Directions,” Sleep and Vigilance, vol. 8, no. 2, pp. 165–167, Nov. 2024, doi: https://doi.org/10.1007/s41782 024 00294 3

[2] M. A. Espinosa, P. Ponce, A. Molina, V. Borja, M. G. Torres, and M. Rojas, “Advancements in Home Based Devices for Detecting Obstructive Sleep Apnea: A Comprehensive Study,” Sensors (Basel, Switzerland), vol. 23, no. 23, p. 9512, Nov. 2023, doi: https://doi.org/10.3390/s23239512

[3] T. Paul, O. Hassan, C. S. McCrae, S. K. Islam, and A. S. M. Mosa, “An Explainable Fusion of ECG and SpO2 Based Models for Real Time Sleep Apnea Detection,” Bioengineering, vol. 12, no. 4, p. 382, Apr. 2025, doi: https://doi.org/10.3390/bioengineering12040382

[4] C. Haberfeld, Alaa Sheta, M. S. Hossain, Hamza Turabieh, and S. Surani, “SAS Mobile Application for Diagnosis of Obstructive Sleep Apnea Utilizing Machine Learning Models,” vol. 18, pp. 0522–0529, Oct. 2020, doi: https://doi.org/10.1109/uemcon51285.2020.9298041

[5] D. Jacob, P. Kokil, S. Subramanian, and T. Jayanthi, “Wearable Sensor Design for Real Time Obstructive Sleep Apnea Monitoring,” IEEE Sensors Journal, pp. 1–1, 2025, doi: https://doi.org/10.1109/jsen.2025.3559296.

[7] M. Talal et al., “Smart Home based IoT for Real time and Secure Remote Health Monitoring of Triage and Priority System using Body Sensors: Multi driven Systematic Review,” Journal of Medical Systems, vol. 43, no. 3, Jan. 2019, doi: https://doi.org/10.1007/s10916 019 1158

[8] D. Peng, L. Sun, Q. Zhou, and Y. Zhang, “AI driven approaches for automatic detection of sleep apnea/hypopnea based on human physiological signals: a review,” Health Information Science and Systems, vol. 13, no. 1, Dec. 2024, doi: https://doi.org/10.1007/s13755 024 00320 8

[9] Ngoc Phu Doan, A. Pham, Hung Manh Pham, Huu Trung Nguyen, Thuy Anh Nguyen, and Huy Hoang Nguyen, “Real time Sleeping Posture Recognition For Smart Hospital Beds,” Oct. 2021, doi: https://doi.org/10.1109/mapr53640.2021.9585289

[10] M. Chowdhury, Golap Kanti Dey, and M. R. Karim, “IEHSAM: IoT based E Health and Sleep Apnea Monitoring System,” May 2021, doi: https://doi.org/10.1109/eti4.051663.2021.9619389

[11] O. Hassan et al., “A Multi Sensor Based Automatic Sleep Apnea Detection System for Adults Using Neural Network Inference on FPGA,” 2022 IEEE International Symposium on Medical Measurements and Applications (MeMeA), pp. 1–6, Jun. 2022, doi: https://doi.org/10.1109/memea54994.2022.9856509

[12] X. Liu and W. Liu, “Innovative Machine Learning Approaches for Real Time Detection of Obstructive Sleep Apnea Using ECG Signals in Medical Wireless Systems,” 2021 Cross Strait Radio Science and Wireless Technology Conference (CSRSWTC), pp. 1–3, Nov. 2024, doi: https://doi.org/10.1109/csrswtc64338.2024.10811486

[13] M. Abuthahir, C. S. Ranganathan, M. Radhika, R Saravanan, J Visumathi, and C. Srinivasan, “Cloud Based SVM Approach for Automated Sleep Stage Prediction in Children Using Wearable Technology,” pp. 1–6, Sep. 2024, doi: https://doi.org/10.1109/acoit62457.2024.10939087

[14] G. Peruzzi, A. Galli, G. Giorgi, and A. Pozzebon, “Sleep Posture Detection via Embedded Machine Learning on a Reduced Set of Pressure Sensors,” Sensors, vol. 25, no. 2, p. 458, Jan. 2025, doi: https://doi.org/10.3390/s25020458

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